

Supplementary material for: A physics-informed digital twin for tube-life-aware operation of industrial steam methane reformers under uncertainty

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S1. Digitisation of the verification curves

Figure 3 of Xu and Froment [2] reports six axial profiles for one industrial reformer tube on three vertical axes: methane and CO₂ conversion on the left axis, total pressure in bar and the process-gas, inner-wall and outer-wall temperatures in kelvin on two right axes. The figure was digitised with WebPlotDigitizer (version 5) from a 600 dpi scan of the printed page. Three independent axis calibrations were defined on the same image, one per vertical axis, each anchored on two gridline labels of the horizontal axis (0 and 10 m) and two labels of the corresponding vertical axis (0 and 0.8 for conversion, 24 and 29 bar for pressure, 800 and 1150 K for temperature). Points were placed manually along each curve at intervals of roughly 0.05 m of the tube length, with additional points concentrated at the discontinuity at the end of the heated length ($z = 11.12$ m), giving 118–261 points per curve.

The left axis of the original figure is labelled $x/(1 + \zeta)$, where $\zeta = 0.056$ is the CO₂/CH₄ molar ratio of the feed; the digitised ordinates of the two conversion curves are therefore multiplied by $1 + \zeta$ to obtain conversions referred to the equivalent methane feed. The residual uncertainty of the digitisation, estimated from the spread of repeated point placements on the gridlines, is about 2 K in temperature, 0.03 bar in pressure and 0.005 in conversion, which is small compared with the model residuals reported in the main text. The digitised curves, the axis calibrations and a tidy merged file with provenance metadata are in `data/literature_validation/digitized/` of the repository.

Two features of the digitised data are recorded but not corrected. The digitised process-gas temperature begins at 764 K whereas Table 2 of the same paper states an inlet temperature of 793 K, so all temperature comparisons in the main text exclude $z < 0.3$ m. Both conversion curves begin at about 0.057 rather than at zero; this value lies between the inlet conversion implied by the higher-alkane rule used here (0.018) and the value obtained if all C₂–C₅ carbon is converted to CO at the inlet (0.091), and the conversion increment $x(z) - x(0.5 \text{ m})$ is therefore used as the primary comparison metric.

S2. Operating space and campaign design

The campaigns evaluated on this space are: an open-loop Latin-hypercube design of 2000 points (surrogate training); a Saltelli design of 4096 points for the Sobol indices and as an independent surrogate test set; a closed-loop 15×15 grid in steam-to-carbon ratio and load at constant outlet temperature (Fig. 5 of the main text); twenty scenario-years of 8760 hourly states; an NSGA-II run of 200 individuals over 300 generations; and a scrambled Sobol Monte Carlo design of 2048 samples propagated through the physics model at eight operating points, with 1024 further samples per uncertainty group for the variance decomposition. All designs use seed 0. The complete pipeline runs in 35 minutes on a ten-core laptop.

Table S1: Operating space sampled in the design, sensitivity, surrogate and optimisation campaigns. The base column is the calibrated Plant A state. Ranges are motivated by industrial practice and by the operating points of the sources used for verification and calibration.

Variable	Unit	Min	Max	Base	Basis
steam-to-carbon ratio	–	2.5	4.0	2.89	carbon-formation limit on Ni catalyst below about 2.5; upper value typical of hydrogen plants; 2.7–3.0 in the reference plant [3], 3.36 in [2]
feed per tube	fraction of base	0.60	1.10	1.00	industrial turndown of about 50–60%; 110 % short-term overload
specific firing $Q_{\text{comb}}/\text{feed}$	fraction of base	0.85	1.15	1.00	burner control range around the plant value
inlet temperature	K	800	900	884.5	884.5 K in the reference plant, 793 K in [2]; cracking in the preheat coil above about 920 K
inlet pressure	bar	25	33	30.1	industrial range 20–40 bar; 28–30 bar in the reference plant, 29 bar in [2]
catalyst activity	–	0.10	0.30	0.20	0.20 fitted to the reference plant for a mid-campaign catalyst [3]; fresh higher, end-of-campaign lower
excess air	%	5	25	21.6	industrial reformer furnaces 5–20 %; upper value extended so that the calibrated base lies inside the sampled space

Table S2: Best surrogate per target ($N_{\text{train}} = 2000$ Latin-hypercube points, $N_{\text{test}} = 4096$ Saltelli points): model family and feature set, test R^2 and RMSE, five-fold cross-validated RMSE, and empirical coverage (PICP) and mean width (MPIW) of the nominal 90 % CV+ conformal interval. “physics” denotes the feature set augmented with the equilibrium methane conversion and the specific duty.

Target	Unit	Model	R^2	RMSE	CV RMSE	PICP ₉₀ / MPIW ₉₀
$T_{\text{wo,max}}$	K	GP, physics	1.000	0.113	0.133	0.957 / 0.25
z/L at $T_{\text{wo,max}}$	–	MLP, base	0.644	0.0319	0.0522	0.971 / 0.050
T_{out}	K	GP, physics	1.000	0.0182	0.0254	0.940 / 0.061
p_{out}	bar	GP, base	1.000	2.87×10^{-4}	5.63×10^{-4}	$0.961 / 1.0 \times 10^{-3}$
CH ₄ slip (dry)	%	GP, physics	1.000	1.25×10^{-3}	1.42×10^{-3}	$0.953 / 4.0 \times 10^{-3}$
H ₂ per tube	kmol h ^{−1}	GP, base	1.000	8.47×10^{-4}	1.16×10^{-3}	$0.946 / 2.9 \times 10^{-3}$
absorbed duty	W	GP, base	1.000	3.9×10^3	4.14×10^3	$0.949 / 1.2 \times 10^4$
$\log_{10} t_r$ at hot spot	–	GP, base	0.9999	5.66×10^{-3}	5.97×10^{-3}	0.959 / 0.011

S3. Surrogate performance

The Gaussian process dominates the other families by one to two orders of magnitude in RMSE on every continuous target, which is expected because the physics model is smooth and deterministic and the surrogate is effectively interpolating it; histogram gradient boosting is the weakest, with an RMSE of about 3 K on the hot-spot temperature. The location of the hot spot is the only target that is not learnable: it is bimodal, sitting near $z/L = 0.43$ for most inputs and moving to the outlet plateau for others, and all three families give R^2 below 0.65. It is reported for completeness and is not used in the life calculation, which requires only the maximum. Prediction takes 38 μs per point against 0.25 s for the physics model.

S4. Steam-credit sweep

S5. Variance decomposition

The one-group-at-a-time shares sum to unity within 0.04, so the four groups act almost additively and the interaction between them is negligible at the base point. The share of the creep group in the hot-spot temperature (0.211) is not a feedback of the creep model on the thermal solution, which does not exist, but the effect of the wall thickness, which belongs to the creep group because it also enters the hoop stress and which changes the conduction drop across the tube wall.

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Table S3: Steam-credit sweep of the iso-production ($H_2 = \text{base} \pm 1\%$) life–fuel front. $\alpha = 1$ counts firebox fuel only; $\alpha = 0$ charges steam raising and feed preheat in full. All rows are physics-verified.

α	Regime	S/C	Load	Firing	Excess air (%)	Rel. life cost	Total fuel vs base (%)	$T_{\text{wo,max}}$ (K)
0.00	min life	3.99	0.96	0.909	24.6	0.059	+3.2	1103
0.00	knee	3.29	0.97	0.850	5.1	0.828	−6.6	1144
0.00	min fuel	2.66	1.00	0.888	5.0	4.539	−8.6	1171
0.25	min life	4.00	0.96	0.943	24.9	0.057	+2.3	1103
0.25	knee	3.80	0.94	0.850	5.0	0.365	−7.1	1131
0.25	min fuel	2.92	0.99	0.865	5.0	2.016	−8.9	1158
0.50	min life	4.00	0.96	1.039	25.0	0.057	+5.5	1103
0.50	knee	4.00	0.95	0.850	13.8	0.103	−7.6	1112
0.50	min fuel	3.63	0.95	0.850	5.0	0.480	−10.5	1135
0.75	min life	4.00	0.96	0.911	25.0	0.057	−6.6	1103
0.75	knee	4.00	0.96	0.850	13.5	0.105	−12.2	1112
0.75	min fuel	4.00	0.93	0.850	5.0	0.276	−14.6	1127
1.00	min life	4.00	0.96	0.958	25.0	0.057	−7.3	1103
1.00	knee	4.00	0.96	0.850	15.3	0.087	−17.7	1109
1.00	min fuel	4.00	0.93	0.850	5.0	0.276	−20.4	1127

Table S4: Variance decomposition at the S1 base point: one-group-at-a-time share of the joint variance ($N = 1024$ per group) and group total-effect Sobol index computed on a Gaussian-process surrogate of the joint sample ($N = 2048$).

Group	share, $\log_{10} t_r$	S_T , $\log_{10} t_r$	share, $T_{\text{wo,max}}$	S_T , $T_{\text{wo,max}}$
kinetics	0.233	0.225	0.513	0.620
heat transfer	0.050	0.114	0.111	0.302
creep	0.622	0.655	0.211	0.093
measurement	0.094	0.036	0.202	0.092
sum	0.999		1.037	

S6. Supplementary figures

References

- [1] J. Xu, G. F. Froment, Methane steam reforming, methanation and water-gas shift: I. Intrinsic kinetics, *AIChE J.* 35 (1989) 88–96.
- [2] J. Xu, G. F. Froment, Methane steam reforming: II. Diffusional limitations and reactor simulation, *AIChE J.* 35 (1989) 97–103.
- [3] D. A. Latham, K. B. McAuley, B. A. Peppley, T. M. Raybold, Mathematical modeling of an industrial steam-methane reformer for on-line deployment, *Fuel Process. Technol.* 92 (2011) 1574–1586.

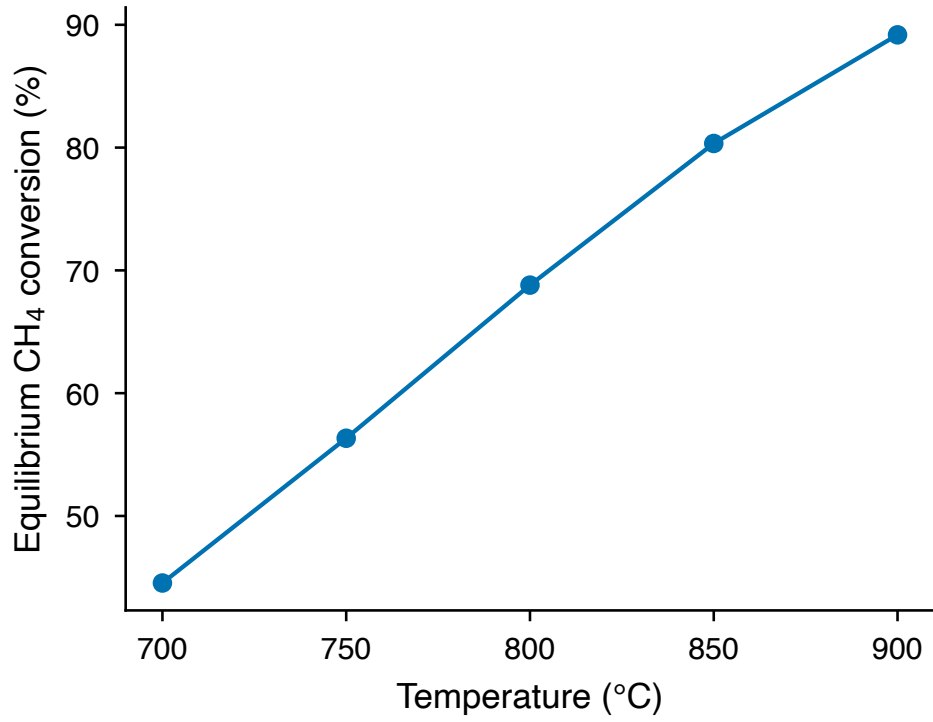


Figure S1: Equilibrium methane conversion of a CH₄/H₂O mixture at 25 bar and a steam-to-carbon ratio of 3, computed with Cantera and the GRI-Mech 3.0 thermodynamic data. This is the thermodynamic ceiling against which the reactor model is checked; at 850 °C the equilibrium conversion is 0.80 and the dry hydrogen fraction 73 %.

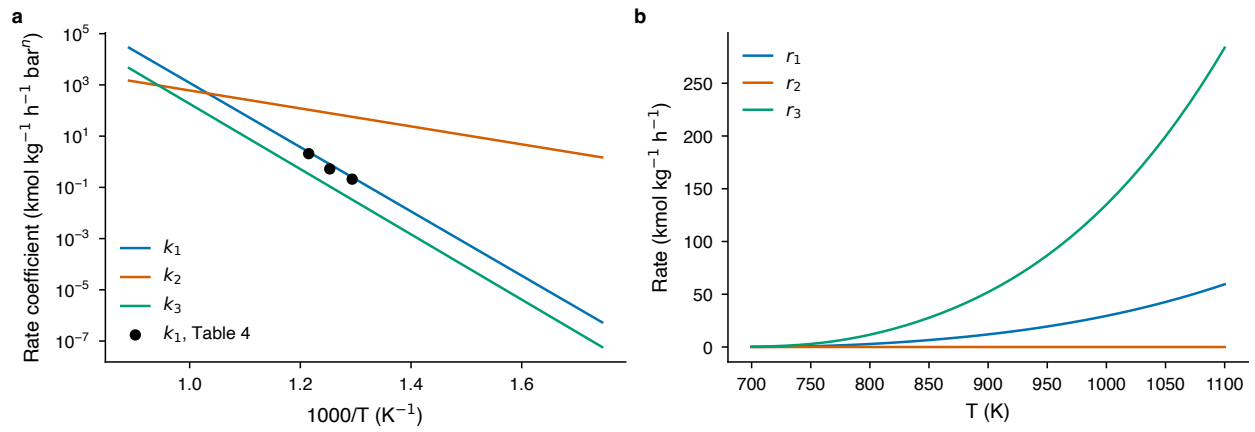


Figure S2: Xu–Froment kinetics as implemented: (a) Arrhenius plot of the rate coefficients k_1 , k_2 and k_3 against $1000/T$, with the per-temperature estimates of Table 4 of [1] overlaid, and (b) the three reaction rates against temperature at 25 bar for a CH₄:H₂O:H₂ feed of 1:3:0.1.

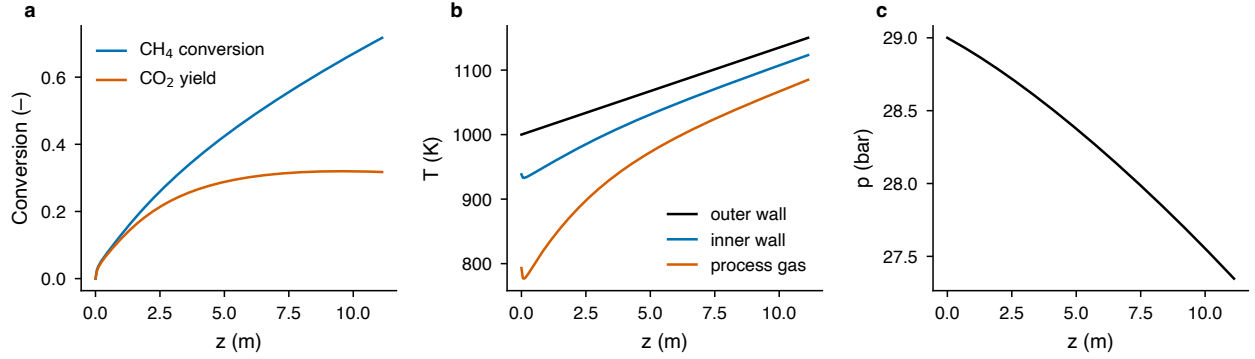


Figure S3: Stand-alone tube model with a prescribed linear outer-wall profile: (a) methane conversion and CO₂ yield, (b) process-gas, inner-wall and outer-wall temperatures and (c) pressure along the tube for the feed of Table 2 of [2]. This run precedes the verification of the main text and uses no fitted parameters.

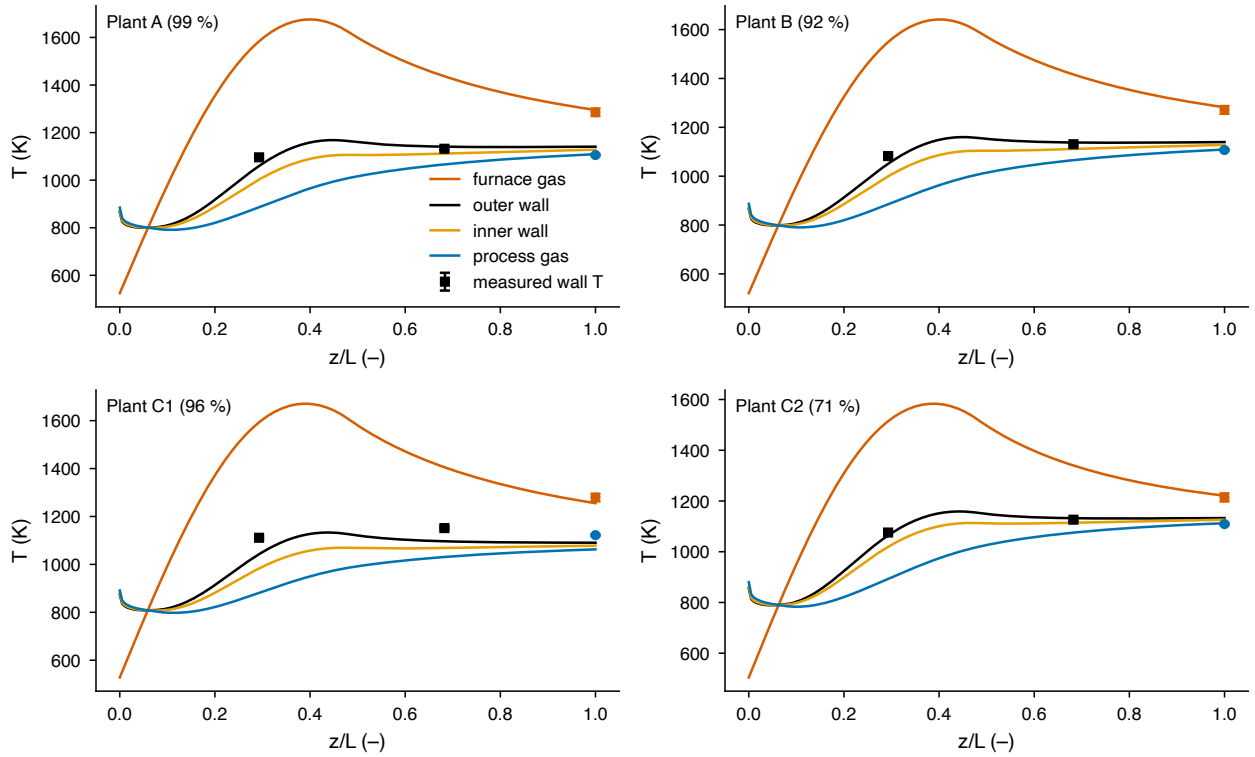


Figure S4: Coupled furnace-tube model before calibration, with the parameter values of Table 6 of [3] transferred directly: axial temperature profiles for the plant cases and the measured wall temperatures. Three of the four cases are reproduced within the measurement uncertainty without any fitting.

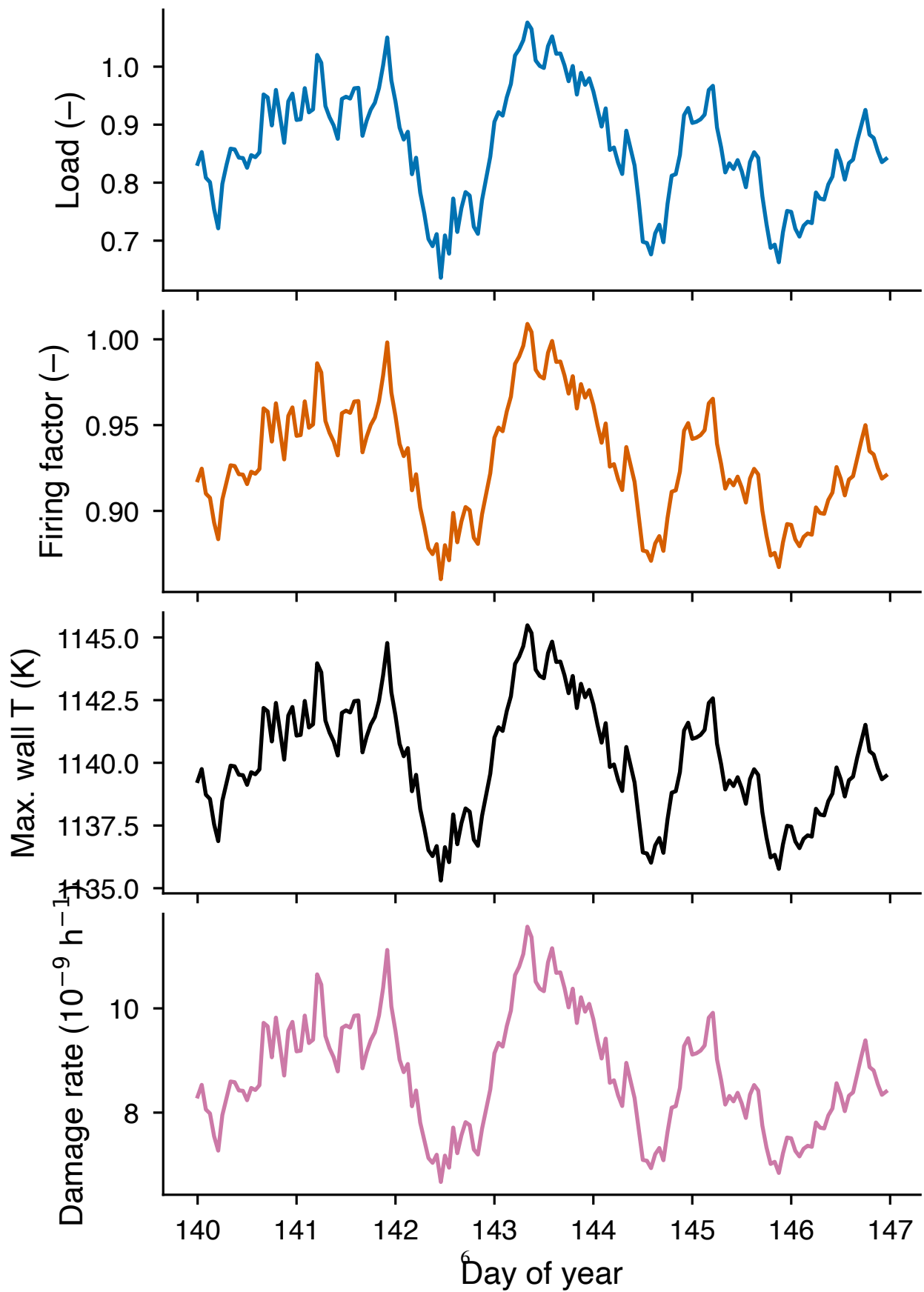


Figure S5: One representative week of the renewable-following scenario S3: load, firing factor, hot-spot temperature and hourly damage rate. The damage rate follows the hot spot through the Larson–Miller relation and is therefore strongly non-linear in the load.